

Luke Serrano

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EXPERIENCE

Land, Air, and Space Robotics (LASR) Lab

Aug 2025 - Present

Computer Vision Lead

College Station, TX

- Lead development and maintenance of computer vision capabilities on robotic testbeds in spacecraft emulation laboratory
- Created an object-detection-based stereo camera relative position tracking algorithm in Robot Operating System (ROS) to support downstream estimation and control
- Developing an optical relative pose estimation algorithm to facilitate precise simulated robotic satellite close-proximity docking and manipulation tasks
- Manage imagery datasets for object detection models, ROS Git repositories, and stereo camera integration on robotic platforms

Sandia National Laboratories Year-Round Intern (Held Clearance)

Jan 2024 - Aug 2025

Missile Guidance, Navigation, and Control (GNC) Engineer

Albuquerque, NM

- Proved ability to work within a multi-language software framework, dive into mathematical theory, develop within a Git workflow, and contribute meaningfully to a GNC team
- Improved navigation regression test framework that stresses the robustness of current and future missile navigation algorithms by implementing the ability to playback flight sensor data
- Analyzed telemetry from IMU calibration procedure and C++ navigation software to create an independent Inertial Navigation System algorithm in an attempt to explain accumulation of velocity-error
- Assisted in the generation of optimal trajectories and creation of a thrust vector control autopilot to supply customer with impact results from an end-to-end Monte Carlo analysis

Missouri S&T Satellite Research Team (M-SAT)

Aug 2022 - May 2025

Co-Chief Engineer and GNC Subsystem Member

Rolla, MO

- Managed Git repositories and NASA's 42 simulator integration for close-proximity GNC team

Personal Projects

- Designed, implemented, and compared EKF/MEKF against Extended Gaussian Mixture Filter for lunar lander pose estimation utilizing inertial and landmark measurements (Python)(Fall 2025)
- Leveraged NASA's Godard Imaging and Analysis Navigation Tool to perform camera calibration, stellar, and relative optical navigation with NASA DAWN imagery (Python)(Spring 2025)
- Implemented an LQR optimal error-quaternion attitude controller for a spacecraft using Colorado Boulder's Basilisk astrodynamics simulation framework (Python)(Spring 2025)

EDUCATION

Texas A&M University

Fall 2025 - Spring 2027

Pursuing M.S. in Aerospace Engineering

GPA: 4.0

Optimal Estimation of Dynamic Systems, Advanced Dynamics, Orbital Mechanics, Optimal Control

Missouri University of Science & Technology

Fall 2022 - Spring 2025

B.S. in Aerospace Engineering

GPA: 3.82

Orbital Mechanics II; Orbit Determination and Estimation; Optimal Control

SKILLS

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|-------------------------------|--|-------------------------|
| - Object-Oriented Programming | - C++ syntax, compilation, and linking | - Git Workflow |
| - Pub/Sub Messaging | - Mathematical Troubleshooting | - Python |
| - Linux Environment | - Simulink Simulation Development | - Self-Directed Learner |